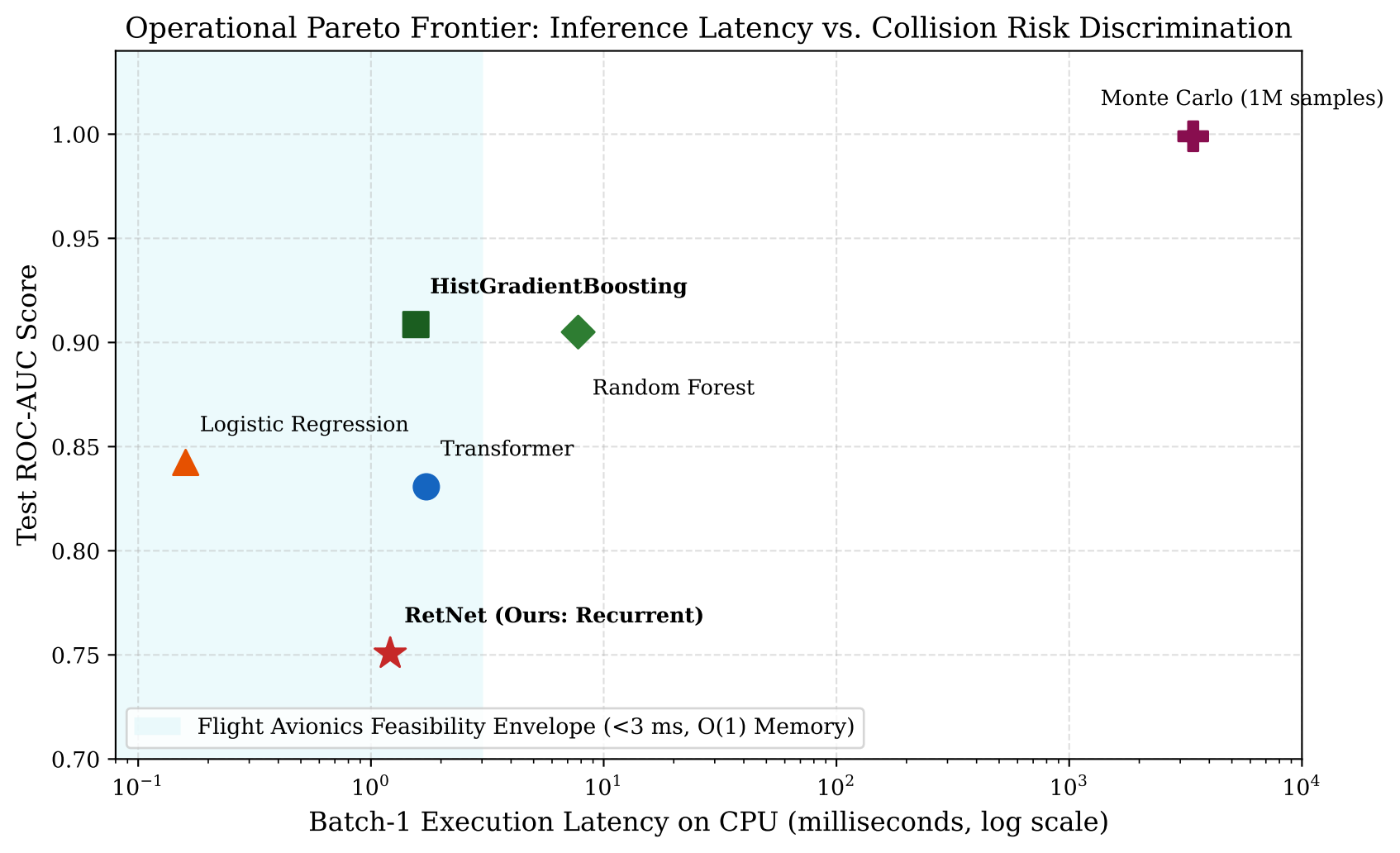
Operational Pareto Frontier: Inference Latency vs. Collision Risk Discrimination
Monte Carlo (1M samples)
HistGradientBoosting
Random Forest
Logistic Regression
Transformer
RetNet (Ours: Recurrent)
Flight Avionics Feasibility Envelope (<3 ms, O(1) Memory)
Test ROC-AUC Score
1.00
0.95
0.90
0.85
0.80
0.75
0.70
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Batch-1 Execution Latency on CPU (milliseconds, log scale)